

Energy Efficient Hardware for Always On Local LLM Server?

Posted by u/iKy1e • 0 points • 59 comments

I have Home Assistant setup controlling most of the things in my house. I can use OpenAI with it to get a custom voice assistant, but I really want a fully local offline setup.

I have played around with different models on my MacBook Pro, and I have a 3080 gaming PC but the laptop isn't a server, and the gaming PC seems way too energy intensive to leave running 24/7.

I'm happy to go by new hardware for this, but if I buy a 4090 and leave it running 24/7 that's up to \$200/month in electrical usage and that's.... too much.

I could go for a raspberry pi and it'd use no power. But I'd like my assistant to respond some time this month.

So I guess my question is: **what's the most energy efficient hardware I can get away with, that'd be able to run say Llama 3 8b in about real time?**

(faster is better, but that's I think about the smallest model and slowest that'd not be painful to use).

Is something like a 4060 energy efficient enough to use for an always on server, and still powerful enough to actually run the models?

Is a Mac mini the best bet? (Mac don't like being servers, auto login, auto boot, network drives unmounting, so I'd prefer to avoid one. But it might be the best option)

Comments

u/ErwinsArm_ • 0 points • 2024-11-15 03:37 UTC

Just rent a vps atp

u/Zyj • 1 points • 2024-07-22 00:29 UTC

Maybe the Snapdragon X Elite is the way to go for you. Either a laptop or the devkit with 32gb RAM

u/iKy1e • 1 points • 2024-07-22 03:01 UTC

> Snapdragon X Elite

I really like the idea of that. My big issue stopping me is wondering what the software support will be like running a dev board for a brand new platform.

u/Zyj • 2 points • 2024-07-22 18:30 UTC

Right! I'm interested in the Dell XPS 13 with the Snapdragon and 64GB RAM but it comes with Windows 11 and i'm not sure how well Linux will run on it.

u/mixed9 • 1 points • 2024-07-23 23:49 UTC

A reason to stick with Mac for now but definitely watching!

u/MoffKalast • 1 points • 2024-07-21 22:15 UTC

The Jetson Orin AGX line is extremely power efficient, but you'll pay for it all upfront, so probably no real gain for the first few decades of running it.